Sample Test Plan
(for Program 2 from last week's Programming Assignment 4)

Rationale for Test Conditions
(Minimal conditions that must be tested in order to validate program correctness)

1. First letter
   a. one invalid first letter
   b. one valid uppercase first letter
   c. one valid lowercase first letter
   d. one test for each case identified by one letter
      i. B
      ii. D
      iii. E

2. Second letter
   a. one invalid second letter after C
   b. one valid uppercase for the second letter after C
   c. one valid lowercase for the second letter after C
   d. one test for each case identified by second letter after C
      i. I
      ii. N
      iii. S
   e. one invalid second letter after I
   f. one valid uppercase for the second letter after I
   g. one valid lowercase for the second letter after I
   h. one test for each case identified by second letter after I
      i. A
      ii. T
   i. one invalid second letter after S
   j. one valid uppercase for the second letter after S
   k. one valid lowercase for the second letter after S

3. Third letter
   a. one invalid third letter after Sx
   b. one valid uppercase for the third letter after Sx
   c. one valid lowercase for the third letter after Sx
   d. one test for each case identified by three letters
      i. SAN
      ii. SEP
      iii. SET

Test cases for program, and which condition will be tested:
Test 1: Will test condition 1a
Test 2: Will test conditions 1b, 1di
Test 3: Will test conditions 1c, 1dii

Test 4: Will test conditions 1c, 1diii
Test 5: Will test conditions 1c, 1a
Test 6: Will test conditions 1c, 2b, 2di
Test 7: Will test conditions 1c, 2c, 2dii
Test 8: Will test conditions 1c, 2c, 2diii
Test 9: Will test conditions 1c, 2e
Test 10: Will test conditions 1c, 2f, 2hi
Test 11: Will test conditions 1c, 2g, 2hii
Test 12: Will test conditions 1b, 2j, 3a
Test 13: Will test conditions 1c, 2i
Test 14: Will test conditions 1c, 2k, 3c, 3di
Test 15: Will test conditions 1c, 2i
Test 16: Will test conditions 1c, 2k, 3c, 3dii
Test 17: Will test conditions 1c, 2j, 3b, 3diii

*Note that between all 17 test cases, **all** test conditions will be tested.*
(If all were not covered, more test cases would be needed)

Test Plan test cases (inputs used and expected outputs):

Test case number	TEST INPUT			EXPECTED OUTPUT	Comments (conditions tested)
	Ltr1	Ltr2	Ltr3	Message	
1	X			Unknown Program	1a
2	B			BS in Business Technology Management	1b, 1d
3	d			MS in Database Technologies	1c, 1d
4	e			Certificate in E-Security	1c, 1d
5	c	x		Unknown Program	1c, 2a
6	c	I		BS in Computer Information Systems	1c, 2b, 2di
7	c	n		BS in Computer Networking	1c, 2c, 2dii
8	c	s		BS in Computer Science	1c, 2c, 2diii
9	i	x		Unknown Program	1c, 2e
10	i	A		MS in Information Assurance	1c, 2f, 2hi
11	i	t		MS in Information Technology Management	1c, 2g, 2hii
12	S	AX		Unknown Program	1b, 2j, 3a
13	s	xn		Unknown Program	1c, 2i
14	s	an		Certificate in Storage Area Networks	1c, 2k, 3c, 3di
15	s	xp		Unknown Program	1c, 2i
16	s	ep		MS in Software Engineering Practices	1c, 2k, 3c, 3dii
17	s	ET		MS in Systems Engineering Technologies	1c, 2j, 3b, 3diii